

Total No. of Questions : 10]

SEAT No. :

P3003

[Total No. of Pages : 3

[5669]-595

T.E. (Information Technology)
HUMAN-COMPUTER INTERACTION
(2015 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8 and Q9 or Q10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Explain any 2 of the following HCI principles in brief. **[5]**

- i) Understand the task
- ii) Reduce Memory Load
- iii) Strive for Consistency
- iv) Prevent Errors/Reversal of Action

b) State and explain UCD principles. **[5]**

OR

Q2) a) What are mental models, and why are they important in interface design? **[5]**

- b) Observe skilled and novice operators in a familiar domain, for example touch and 'hunt-and-peck' typists, expert and novice game players, or expert and novice users of a computer application. **[5]**

What differences can you discern between their behaviors?

Q3) a) How does making a call differs when using **[5]**

- i) Cell phone
- ii) Smart phone?

Consider the kinds of user, type of activity and context of use.

- b) What can a system designer do to minimize the memory load of the user? **[5]**

P.T.O.

OR

- Q4)** a) Suggest ideas for an interface which uses the properties of sound effectively? [5]
b) When systems are not designed to match the way people actually work, then users end up having to do 'work around'. Discuss. [5]

- Q5)** a) What is design? What is the golden rule of design? Illustrate the process of interaction design. [8]
b) Ascenario is an idealized but detailed description of a specific instance of human-computer interaction (HCI). Scenarios specify how users carry out their tasks in a specified context. Write scenarios for purchasing an airline ticket. [8]

Note : Generate scenarios to cover a wide range of situations, not just the most common ones. Include problem situations that will test the system concept, not just straightforward scenarios.

OR

- Q6)** a) Explain HCI design process with neat diagram. [8]
b) What is a prototype? Explain different types of rapid prototyping techniques. [8]
- Q7)** a) Explain Prototyping with hill-climbing approach. [8]
b) Explain Nielsen's ten heuristics. [8]

OR

- Q8)** a) Discuss Shneiderman's eight golden rules of interface design with suitable examples. [8]
b) Write short note on : [8]
i) Wire-Framing.
ii) Model-View-Controller (MVC) Framework.
- Q9)** a) Complete the cognitive walk-through example for the video remote control design. [9]
b) Discuss applications meant for computer mediated communication. [9]

OR

Q10) a) In order to clean the house

[9]

1. get the vacuum cleaner out
2. fix the appropriate attachment
3. clean the rooms
 - 3.1 clean the hall
 - 3.2 clean the living rooms
 - 3.3 clean the bedrooms
4. empty the dust bag
5. put the vacuum cleaner and attachments away

Plan 0: do 1 – 2 – 3 – 5 in that order
when the dust bag gets full do 4

Plan 1: do any of 3.1, 3.2 or 3.3 in any order
depending on which rooms need cleaning.

For this HTA description of vacuum cleaning, present the same information in a diagrammatic form.

- b) Consider the activity of making a telephone call. Record the actions in an HTA diagram or textually. Start off simply, assuming you know the number to dial, but then add more complicated situations : finding the number in an address book, or what to do when the number is engaged.

[9]
